

Data Publication

Introduction to Data Management Practices course

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Why submit to a repository?

“The data is available upon request”

Many reasons:

- Open Science & FAIR
- Reproducibility
- Trail of evidence
- 3rd party access
- Archival purposes
- Publication of paper requires it



Digitalbevaring.dk

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Why submit to a repository?

Data publication is the best way to make your research projects FAIR since your data becomes:

- **Findable** by being assigned a persistent identifier, and by being described with rich metadata
- **Accessible** by being put in a resource that is searchable, and enables easy access via internet
- **Interoperable** by using standard format and language to represent both the data and its metadata
- **Reusable** by fulfilling the F, A, and I, and by having a clear and accessible data usage license

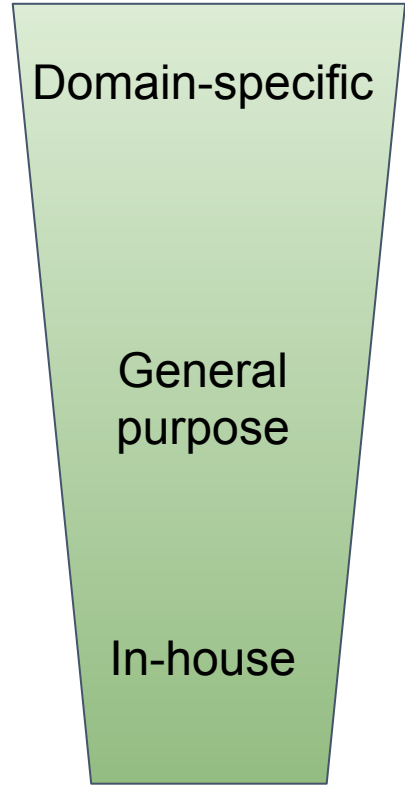
What is data?

What research outputs should be submitted?

- Raw data: straight from the instrument eg fastq, bam, cram
- Processed data: normalization, removal of outliers, expression measurements, statistics
- Metadata: minimum information to reproduce the data, sample information, precise protocols

Types of repositories

- Domain-specific:
 - Best choice - long-term plan, typically free, maximum reach
 - E.g. [European Nucleotide Archive](#), [European Genome Phenome Archive](#), [ArrayExpress](#), [PRIDE](#)
- General purpose:
 - Second best - long-term plan, might cost (now or in future), good reach but less specific in metadata → more difficult for future users to judge if a dataset will be useful
 - E.g. [Zenodo](#), [\(SciLifeLab\) Figshare](#), [Dryad](#)
- In-house/institutional
 - For archive/backup purpose mainly, might cost, limited reach unless also published in a data catalogue



Evaluate a repository

Things to check when evaluating:

- Are others in the community using it?
- Is it easy to navigate / user-friendly?
- Is there support / guidance for submission and reuse?
- Is it sustainable, i.e. will the repository be around for a while?
- Will the datasets obtain persistent identifiers? Is the repository itself FAIR?

Identify repositories

How to find a suitable repository for life science data?

- [EBI repository wizard](#) - guide depending on data type
- [ELIXIR deposition databases](#) - core resources with long-term data preservation and accessibility plans
- [FAIRsharing.org/databases](#) - catalogue of many repositories, with possibility to filter on e.g. domain
- [Scientific Data Repository Guidance](#) - publisher's recommendation
- [re3data.org](#) - registry of research data repositories (not only life science)

Demo: EBI Repository Wizard

Which repository would be suitable if you have a genomics project with mice RNA sequences?

- Go to <https://www.ebi.ac.uk/submission/>
- Answer the questions regarding
 - data type (DNA/RNA sequence)
 - need for controlled access (No)
 - if experimentally produced by you (Yes)
 - type of study (Other)
- Solution: [European Nucleotide Archive \(ENA\)](#)

Key Points

- Publishing data greatly increases the FAIRness of your research.
- Benefits of sharing data are several e.g. reproducibility purposes, follow the Open Science directive, meet requirement from publishers.
- If possible, use a domain-specific repository since it has maximum reach in the research community.
- The research output data types determines which domain-specific repository is suitable.